



Immediate Effects of Endurance Exercise on Subsequent Strength Performance

10

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The Acute and Chronic Hypotheses

The order in which strength and endurance training are performed within a concurrent training regimen can vary. Previous work has employed strength training prior [1–3] and subsequent to [4–7] endurance exercise. As the focus of much work into concurrent training is the inhibition of strength development, it is important to understand the influence of any prior endurance exercise on strength-type performance.

Within the concurrent training paradigm there exist the acute and chronic hypotheses pertaining to the “interference effect”. The chronic hypothesis suggests that the trained muscle is placed under conflicting training stimuli during a longitudinal concurrent training programme, as the muscle is attempting to adapt simultaneously to both strength and endurance training [8]. This chapter will however not focus on the chronic effect, but rather the acute effect of endurance exercise on subsequent strength performance. The acute hypothesis, that was initially proposed by Craig et al. [9], relates to the inhibition of strength development resulting from endurance (in this case running) training carried out immediately prior to the strength training. In other words, simply performing strength and endurance training concurrently may not inhibit strength development; rather, that endurance loading immediately prior to strength training results in diminished quality, volume, and intensity of strength training due to residual fatigue. It was proposed that this residual fatigue resulting from prior endurance training might compromise the ability of the trained muscles to develop adequate muscular tension during strength training [9]. Consequently, the ability of muscle to generate a sufficient stimulus to complete the strength session could be compromised and hence reduce the potential for a positive

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adaptive response. Despite the acute hypothesis being related to acute concurrent training, it still has relevance in longitudinal training scenarios. If strength performance is repeatedly reduced as a result of prior endurance training, it is likely that the magnitude of strength development will be inhibited [1, 2] when compared with strength training performed in isolation.

The acute hypothesis, like many aspects of concurrent training, is equivocal. Interestingly, much of the research reported reduced strength performance, whereas some other studies (3 out of 25 studies) reported that a prior bout of endurance training does not affect strength performance (summary presented in Table 10.1). As such, it is conceptually possible that any negative effect of prior endurance activity on subsequent strength performance may be dependent on modality, intensity, and duration of both the endurance and strength training.

Endurance Exercise-Induced Fatigue

Fatigue is an extremely broad term and is attributable to numerous physiological processes. Fatigue mechanisms can generally be subdivided into central and peripheral origins. Peripheral fatigue results from impaired processes at, or distal to the neuromuscular junction, whereas central fatigue is caused by impaired processes within the central nervous system (CNS) [38] (Fig. 10.1). In applied sport and exercise science, fatigue can be defined as an exercise-induced impairment in the ability to produce muscular force [16, 38] and sustain increased force in the presence of an increased perception of effort [39]. The fatiguing effect of endurance-type exercise has received substantial attention in scientific literature and is well documented [33, 40–44] and there is a great deal of research available pertaining to the effect of endurance loading on subsequent performance in an acute setting [8, 15, 17, 24, 31, 34, 37].

It has been frequently reported that central and/or peripheral fatigue following endurance exercise cause acute declines in contractile strength (Table 10.1). Previous research has also indicated that the contributions of central and peripheral fatigue are largely dependent on the volume and intensity of exercise performed. For example, during sustained higher intensity contractions of an isolated muscle group, peripheral fatigue appears to be the dominant factor contributing to fatigue. However, the contribution from central mechanisms increases as the exercise task increases in duration [16, 45, 46]. Consequently, it is reasonable to suggest that it is the central, rather than peripheral, factors that contribute to fatigue and hence to the decrement in strength performance that is often observed when endurance exercise acutely precedes strength training. However, it is also plausible that peripheral mechanisms make some contribution to any reduction of strength performance in the short term following higher intensity lower volume endurance loading. Consequently, any impaired strength performance and contributions from central and/or peripheral fatigue appear to be dependent on the endurance exercise and the nature of subsequent strength session and recovery (or “relief period”) between endurance and strength stimuli.

Table 10.1 Summary of research that assessed strength performance following endurance exercise

Reference	Subjects ^a	END modality	END volume and intensity	Relief duration	ST modality	ST loading/ assessment	Impact of END loading on strength performance
Jones et al. [10]	Strength-trained males	Treadmill running	30 min 70% $\dot{V} O_{2max}$	None	Back squat Deadlift Bench press Prone row Military press	5 × 6 80% 1RM	↓
Thomas et al. [11]	Male semi- professional soccer players	Simulated soccer match	90 min	None 24 h 48 h 72 h	Knee extension	MVC	↓
Eklund et al. [7]	Recreationally active females	Cycle ergometry	30 min 65% W_{max}	None	Leg press	1RM	↓
García- Pinillos et al. [12]	Trained endurance runners (males and females)	Running	MSFT	None	CMJ	Max effort	↑
Jones et al. [13]	Strength-trained males	Cycle ergometry	30 min 70% $p \dot{V} O_{2max}$	10 min	Leg press Knee extension	5 × 6 80% 1RM	=
García- Pinillos et al. [14]	Experienced recreational male long-distance runners	Running on track	4 × 3 × 400 m	None	CMJ	Max effort	↑
Panissa et al. [15]	Physically active males	Treadmill running Cycle ergometry	15 × 1 min 50% I_{max}	None	Half squat on smith machine	4 sets to failure 80% 1RM	↓

(continued)

Table 10.1 (continued)

Reference	Subjects ^a	END modality	END volume and intensity	Relief duration	ST modality	ST loading/ assessment	Impact of END loading on strength performance
Thomas et al. [16]	Well-trained male cyclists	Cycle ergometry	Time trial of; 4 km 20 km 40 km	None	Knee extension	MVC	↓
De Salles Paimelli et al. [17]	Recreationally strength-trained males	Treadmill running	5 km continuous 90% anaerobic threshold velocity or 5 km intermittent 1:1 min work periods $\dot{V} O_{2max}$	10 min	Leg press Bench press	1 RM 4 sets to failure 80% 1RM	↓
Latorre-Román et al. [18]	Sub-elite male long-distance runners	Running on track	4 × 3 × 400 m	None	CMJ	Max effort	↑
Schumann et al. [19]	Physically active males	Cycle ergometry	30 min 65% MAP	None 24 h 48 h	Isometric leg press	MVC	↓
Taipale et al. [20]	Recreationally endurance trained males and females	Running on track	60 min 80% $\dot{V} O_{2max}$	None	Isometric leg press CMJ	MVC Max effort	↓
Tan et al. [21]	Strength-trained males	Elliptical ergometer (lower limbs only)	30 min 70% age predicted HR_{max}	10 min	Back squat Bench press	3 sets to failure 75% 1RM	↓
Reed et al. [22]	Strength-trained males	Cycle ergometry	45 min 75% HR_{max}	None	Back squat Bench press	6 sets to failure 80% 1RM	↓

Taipale and Häkkinen [23]	Recreationally endurance trained males and females	Running on track	60 min between LT and RCT	None	Isometric leg press	MVC RFD	↓
Panissa et al. [24]	Variety of athletes	Treadmill running	5 km intermittent 1:1 min v $\dot{V}O_{2max}$	30 min 60 min 4 h 8 h 24 h	Half squat on smith machine	4 sets to failure 80% 1RM	↓
Boullosa et al. [25]	Male and female experienced endurance athletes	Running on track	Incremental protocol to exhaustion (mean speed = 18.9 km h ⁻¹ , mean time = 1476 s)	None	CMJ	Max effort	↑
de Araujo Ruas et al. [26]	Strength-trained males	Cycle ergometry	30 min OBLA	None	Leg press	10RM To failure	↓
Boullosa and Tuimil [27]	Male distance athletes	Running on track	Incremental protocol to exhaustion (mean speed = 20.8 km h ⁻¹ , mean time = 322 s)	None	CMJ	Max effort	↑
Lemos et al. [28]	Physically active elderly women	Treadmill running	20 min 60% HR _{max} or 20 min 80% HR _{max}	2 min	Leg press Knee extension Leg curl	10RM To failure	↓
de Souza et al. [29]	Physically active males	Treadmill running	5 km continuous anaerobic threshold velocity or 5 km 1:1 min v $\dot{V}O_{2max}$	10 min	Inclined leg press Bench press	80% 1RM To failure	↓

(continued)

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Reference	Subjects ^a	END modality	END volume and intensity	Relief duration	ST modality	ST loading/assessment	Impact of END loading on strength performance
Vurimaa et al. [30]	Male long and middle distance athletes	Treadmill running	Maximal run: Incremental to exhaustion Tempo run: 40 min at 80% $\dot{V} O_{2max}$ Intermittent run: 40 min 2 min on, 2 min off at 100% $\dot{V} O_{2max}$	None	CMJ Half squat	Max effort 10 reps at 35% 1RM	↑
Goto et al. [5]	Recreationally strength-trained males	Cycle ergometry	60 min 50% $\dot{V} O_{2max}$	15 min	Isometric knee extension	MVC	=
Sporer & Wenger [31]	Active males	Cycle ergometry	6 × 3 min 95–100% $p \dot{V} O_{2max}$ or 36 min 70% $p \dot{V} O_{2max}$	4 h 8 h 24 h	Leg press Bench press	4 sets to failure 75% 1RM	↓
Bentley et al. [32]	Trained male cyclists	Cycle ergometry	30 min 80% $\dot{V} O_{2max}$	None 6 h	Isometric knee extension	MVC	↓
Lepers et al. [33]	Trained male cyclists	Cycle ergometry	2 h 65% MAP	None	Knee extension	MVC	↓
Leveritt et al. [34]	Strength-trained males	Cycle ergometry	50 min 70–100% critical power	8 h 32 h	Isokinetic Isometric Isotonic knee extension	5 MVC between 1.04 and 8.37 rad s ⁻¹	=

Leveritt and Abemethy [8]	Recreationally active males and 1 female	Cycle ergometry	5 min intervals 40, 60, 80% peak $\dot{V}O_2$ 2 min peak $\dot{V}O_2$	30 min	Isokinetic squats Isokinetic knee extension	3 MVC 5 MVC	↓
Bentley et al. [35]	Male-trained cyclists	Cycle ergometry	30 min Lactate threshold	6 h 24 h	Isokinetic knee extension 60° s^{-1} 120° s^{-1} 180° s^{-1} concentric squat jump	MVC	↓
Abemethy et al. [36]	Active males and females	Cycle ergometry	150 min $35\% p \dot{V}O_{2\max}$ or $5 \times 5 \text{ min } 40, 60, 80 \text{ and } 100\% p \dot{V}O_{2\max}$	60 min 4 h	Isokinetic knee extension	MVC between 0.52 and 5.20 rad s^{-1}	↓
Jacobs et al. [37]	Male athletes (runners) and physical education students	Treadmill running Cycle ergometry	Treadmill running: 75 min Max velocity that could be maintained Cycle ergometry: 30 min $70\% \dot{V}O_{2\max}$	1 h 2 h	Isokinetic knee extension	50 MVC	↓

^aAs described in individual studies, *CMJ* countermovement jump, *END* endurance training, *HR*_{max} maximum heart rate, *I*_{max} maximal intensity attained on incremental test, *LT* lactate threshold, *MAP* maximal aerobic power, *MSFT* multistage fitness test *MVC* maximal voluntary contraction, *OBLA* onset of blood lactate accumulation, *p* $\dot{V}O_{2\max}$, power output at $\dot{V}O_{2\max}$; *RCT* respiratory compensation threshold, *RFD* rate of force development, *RM* repetition maximum, *ST* strength training, *TT* time trial, *v* $\dot{V}O_{2\max}$, velocity at $\dot{V}O_{2\max}$; *W*_{max} maximal workload

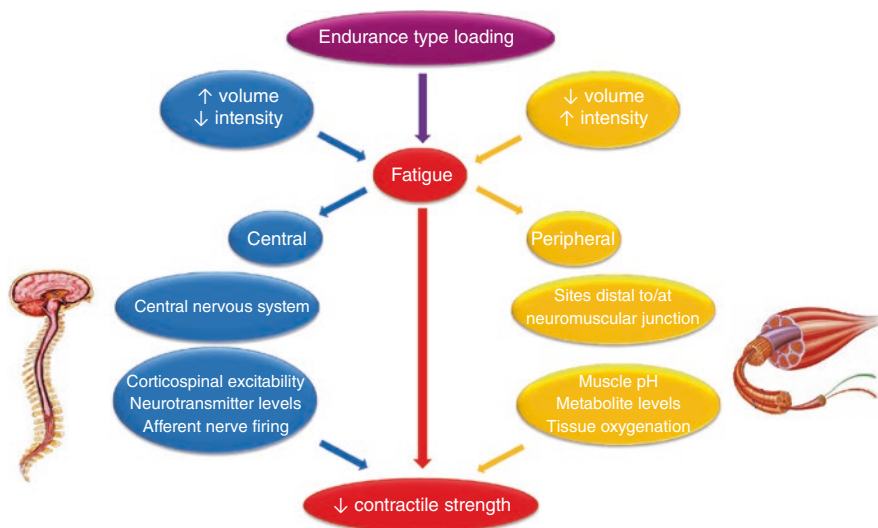


Fig. 10.1 Schematic representation of central and peripheral fatigue mechanisms

Relief Period Between Endurance and Strength Loadings

The “relief period” refers to the time period between endurance and strength loadings. In the available body of peer reviewed literature, relief periods range from strength performance being assessed immediately following endurance loading to assessments conducted 72 h post-endurance loading. It appears that the duration of the relief period may influence the presence of any impairment in strength performance following endurance loading.

Reduced strength performance is observed when the relief between endurance and strength training is shorter (≤ 60 min; Table 10.1). Additionally, reduced strength performance has been observed following longer relief periods of 2, 4, 6, and 8 h [31, 32, 36, 37]; in fact, impaired maximal voluntary contractions (MVC) have also been reported following relief periods of 24, 48, and 72 h [11]. This latter impairment [18], however was observed following a 90-min simulated soccer match (where the exercise stimulus is a combination of both metabolic and mechanical stressors) and represents a very different stimulus to other research. Other studies tend to employ volumes of between 30 and 60 min and intensities of 70–80% heart rate max (HR_{max}), $\dot{V}O_{2max}$, or similar (Table 10.1). Studies which assess strength performance following relief durations of < 8 h and ≥ 8 h only report strength performance to be decremented within 8 h of endurance exercise [19, 24, 31, 35]. In addition, elite kayakers have been reported to separate strength and endurance training sessions by 6–8 h to allow full glycogen restoration [1], which resulted in greater strength increases than those who performed endurance and strength training in closer proximity. Collectively, these data indicate that impairments in strength

performance are likely for ≤ 8 h following endurance loading. As such, if the primary aim of the session is for the individual to perform high quality strength training, in which the neuromuscular system is challenged by high force and/or velocity-based movements the scheduling of strength and endurance stimuli should be considered. If strength training must be conducted in close proximity to endurance training, it is perhaps more appropriate for the session to focus on fibre hypertrophy or strength endurance. Additional caveats to these considerations include the stage of the periodised plan and the training status of the individual. For instance, if the session takes place early in an individual's periodised plan or the individual has limited strength training experience and is unable to produce high levels of contractile force, the aforementioned considerations need not be made.

Modality, Volume, and Intensity of Endurance and Strength Loading

As previously stated, the research tends to employ endurance volumes of between 30 and 60 min at intensities of 70–80% HR_{max} , $\dot{V}\text{O}_{2\text{max}}$, or similar, prior to strength training. The primary endurance training modality employed in these studies is cycle ergometry [7, 8, 15, 16, 19, 22, 26, 31–33, 35–37] with some others [10, 15, 17, 20, 23, 24, 28, 29, 37] employing running (Table 10.1). Thus, at face value, the mode of endurance exercise does not determine the impaired strength performance. This is perhaps surprising as a meta-analysis has indicated that concurrent training with running, but not with cycling results in the inhibition of lower body strength development [47]. It was proposed that this may be attributable to the greater eccentric loading associated with running and possibly the fact cycling involves biomechanical movement patterns with greater similarity to many of the compound lifts employed as measures of lower body strength [48, 49]. However, as previously stated the modality of endurance loading does not appear to influence subsequent strength performance in an acute setting.

Although the presence of any reduced strength performance following endurance exercise appears to be independent of the modality of endurance stimulus, the method and muscle group in which strength is assessed does influence strength impairment. Upper body strength may be unaffected by prior primarily lower-body endurance loading; previous work has reported decreased lower, but not upper, body strength following lower limb endurance activity such as running [17, 29], elliptical training [21], and cycling [22, 31]. Consequently, for such impaired strength performance to occur, the endurance exercise needs to be body part specific; insofar as lower body endurance exercise might influence subsequent lower body strength, but not upper body strength performance. As such, it may be suggested that endurance training modalities such as rowing and arm crank cycling may result in reduced upper body strength performance.

Decrement in strength performance has been observed following 15 [15], 20 [28], 30 [7, 10, 19, 21, 26, 32, 35], 36 [31], 45 [22], 60 [20, 23], 75 [37], 90 [11], 120 [33], and 150 min [36] of endurance loading. As such, the volume of endurance

loading does not appear to be a primary determinant for subsequent impairment of strength performance. However, it is possible that the volume of endurance exercise influences the recovery time course of strength performance. Following 90 min of a simulated soccer game, strength (as assessed via knee extension MVC) performance has been reported to be impaired for up to 72 h [11]. As previously stated, following shorter durations of endurance loading strength performance typically returns to “baseline” within 8 h of endurance loading [19, 24, 31, 35]. However, it is also likely that the intensity of the simulated soccer match was a higher intensity than the other endurance exercise.

Few studies directly compared intensities of endurance exercises on subsequent strength-type performance. Lemos et al. [28] observed greater impairments in leg press, knee extension, and leg curl 10 repetition maximum (RM) reps to failure following 20 min of treadmill running at 80% HR_{max} than the same volume of running at 60% HR_{max} . Whilst this may indicate that higher intensities of endurance exercise can result in greater reduction of strength performance, it should be noted that this observation was made exclusively in a physically active yet elderly female population. Different responses may have been observed in a more highly trained cohort whom are more accustomed to higher intensity endurance loading, thus a reduction in strength is perhaps less likely. During self-paced endurance exercise, the intensity and volume of the exercise may influence the contributions of central and peripheral fatigue. Following a higher intensity, shorter duration 4 km time trial (TT) impaired strength and peripheral dominant fatigue mechanisms have been observed. Following higher volume and lower intensity TTs of 20 and 40 km impaired strength was still observed yet central fatigue mechanisms were dominant [16]. This finding may suggest an intensity and volume-dependent influence on the neuromuscular underpinnings to fatigue and impaired strength performance following endurance exercise.

Direct comparisons are available between continuous and intermittent endurance exercise [17, 29, 31, 36] and subsequent strength performance. This line of work has yielded equivocal findings, with impaired strength performance being observed following both continuous and intermittent endurance exercises [31, 36], intermittent but not continuous endurance exercise [29] and vice versa [17]. As such, the influence of it may be suggested that it is neither the continuous nor intermittent nature of the endurance exercise, which influences any impaired strength performance. The available evidence based suggests that mode of exercise (intermittent and continuous) is not the determining factor, but rather that engaging in endurance exercise per se results in potential detriments in strength performance. This suggestion is supported by the fact that both de Salles Painelli et al. [17] and de Souza et al. [29] employed identical intermittent and continuous endurance exercises in comparable cohorts, yet de Souza et al. [29] observed strength performance only to be reduced following intermittent endurance exercise, whereas de Salles Painelli et al. [17] reported leg press reps to failure only to be lower following continuous endurance exercise.

As previously discussed, various relief periods, modalities, intensities, and volumes of endurance exercise have been employed in research (Table 10.1). Although some of the aforementioned programme variables appear to influence strength performance following endurance exercise (particularly the relief period), it is also

important to note the method in which strength is assessed can have a noteworthy influence on how strength-type performance is interpreted. Many studies assess subsequent strength performance via reps to failure at a specified relative load [15, 17, 21, 22, 24, 26, 28, 29, 31], others employ more high-intensity measures such as 1–5 RM/MVC [5, 8, 11, 16, 17, 19, 20, 23, 32–36, 50] and few utilise more “realistic” strength training protocols [10, 13]. These measures represent different strength qualities, with reps to failure being indicative of strength endurance, and 1–5 RM/MVC representing maximal strength. Nonetheless, inhibited strength endurance, maximal strength both occur following endurance exercise (Table 10.1). Few studies employed measures of both strength endurance and maximal strength following the same endurance and strength loading protocols. As such, it is difficult to make definitive statements regarding which quality is more susceptible to endurance-induced fatigue. Previously, work has indicated that more explosive, “power”, and/or velocity-based phenotypes are more susceptible to endurance exercise-induced decrements in performance than maximal strength and strength endurance indices [9, 51–53]. This finding is, at least in part, supported by observations that impairments in rate of force development (RFD) in the isometric leg press [23] and countermovement jump height (CMJ) [20] following 60 min of running. As previously stated, upper body strength training following endurance exercise (primarily involving the lower body musculature) remained unimpaired [17, 21, 22, 29, 31]. This may indicate peripheral fatigue at, or distal to the neuromuscular junction, to be a major contributing factor to any decrements in lower body strength performance. Based on these observations, it can be recommended that high velocity and explosive lower body phenotypes should be trained independently from any endurance stimulus.

There is limited evidence to suggest that in acute settings strength performance is not inhibited by prior endurance exercise. Research in which strength performance remains unimpaired post-endurance exercise has assessed strength performance via isolated limb actions such as knee extension [5, 13, 34]. However, in the larger body of evidence, strength performance is reduced following endurance exercise when assessed via multi joint exercises such as squats [8, 10, 15, 21, 22, 24] or leg press [7, 17, 19, 20, 23, 26, 28, 29, 31], which also has a greater applicability to athletes, coaches, and practitioners. Therefore, it is reasonable to suggest that multi joint movements involving large amounts of muscle mass are more susceptible to endurance exercise-induced fatigue than isolated limb single joint movements.

Post-activation Potentiation

Notwithstanding the previous issues relating to strength performance, there is some suggestion that performing an endurance-type stimulus can have a potentiating effect and thus a positive acute effect on strength performance [12, 14, 18, 25, 27, 30]. However, this has been reported almost exclusively in trained endurance athletes. Consequently, it appears that those accustomed to endurance loading are perhaps more fatigue resistant to endurance exercise than strength-trained individuals in the acute period following endurance stimulus. Furthermore, in the case of highly

endurance-trained individuals, a prior bout of endurance loading acts as a warm up or “primer” resulting in improved ability to generate force; the same is not true for those whom are not highly endurance trained.

Summary

The relief duration between endurance and strength-type loadings appears to be the primary determinant for any decrement in strength performance. Typically, acute strength performance returns to baseline ≥ 8 h following endurance exercise and therefore if strength sessions follow endurance, then a period of relief of not less than 8 h seems pragmatic to optimise the performance in the strength training.

Reduced strength performance appears to be independent of the endurance modality (cycling or running) and of a continuous or intermittent nature. Impaired strength performance is observed following both; lower volume higher intensity and higher volume lower intensity endurance exercise. Whilst the presence of decremented strength performance appears to be independent of volume and intensity of endurance exercise, the contributions of central and peripheral fatigue are not. Peripheral fatigue mechanisms are dominant following lower volume higher intensity endurance exercise and central fatigue mechanisms are dominant following higher volume lower intensity endurance exercise.

Upper body strength remains unimpaired following lower body endurance exercise. Lower body multi joint movements involving higher muscle mass and velocity based/explosive movements are likely the most susceptible to endurance exercise-induced impairments in strength-type performance (Fig. 10.2).

Implications for Programming

Whilst not recommended practice, due to logistical and scheduling constraints both elite athletes and recreational exercisers will perform strength training subsequent, and in close proximity to, endurance exercise. As detailed in this chapter, this structure will likely lead to some degree of impaired strength performance, particularly when conducted within 8 h of endurance exercise and during lower body lifts that require the engagement of larger volumes of muscle mass. Therefore, if high quality strength training and strength development are the primary aims of the training intervention or macro/meso/micro cycle practitioners seeking to optimise performance during strength training should consider the following:

- Where possible perform strength training ≥ 8 h after any endurance exercise.
- If strength training must be performed in close proximity to lower body endurance exercise, the session should focus on upper body strength development.
- If strength training must be performed in close proximity to lower body endurance exercise, the session should avoid lower body explosive and velocity-based movements and multi joint lifts involving large amounts of muscle mass.

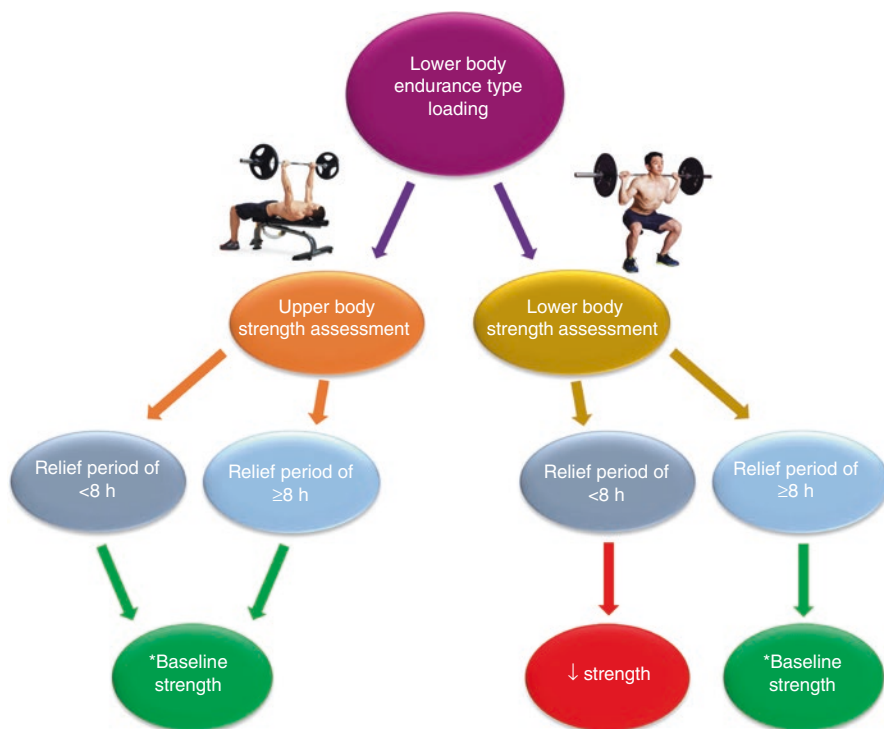


Fig. 10.2 Schematic representation of the “typical” influence of prior endurance loading on subsequent strength performance. Previous work detailed throughout this chapter has indicated that the method by which strength performance is assessed and the relief period are the primary determinants of acute strength performance following endurance loading. *Baseline = prior any endurance loading

It should be noted that it is likely not feasible to implement these evidence-based recommendations year-round, nor is it necessary to. However, during pre-season or pre-competition periods in which practitioners and athletes may be seeking to optimise longer term athletic strength development or enhance velocity-based phenotypes the recommendations here should be considered.

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